

Pranav Kairon

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Summary

I specialize in the application and theoretical exploration of advanced computational tools such as quantum Monte Carlo, tensor networks, and machine learning, with a focus on their applications in quantum computing, condensed matter physics and non-equilibrium dynamics. Additionally, I plan to work on quantum machine learning and obstacles therein.

Education

University of Illinois Urbana Champaign

PHD IN PHYSICS

Urbana, USA

August 2024-Present

- Working with Prof. Bryan Clark
- Teaching Assistant for PHYS 211 (Classical Mechanics)
- CGPA: 4.0/4

University of British Columbia

M.S. IN PHYSICS

Vancouver, Canada

2021 – August 2024

- Thesis: **From quantum computing to condensed matter : a machine learning perspective**
- Teaching Assistant for PHYS 100 (Classical Mechanics) and PHYS 131 (Electromagnetism).
- CGPA: 3.65/4.

Delhi Technological University

B.TECH. IN ENGINEERING PHYSICS

New Delhi, India

2017 – 2021

- Thesis: **Coherence-based inequality for state discrimination**
- CGPA: 8.94/10.

St. Xavier's Senior Secondary School

GRADE 12 (SENIOR SECONDARY LEVEL)

New Delhi, India

2017

- Percentage: 89.2%.

Publications and Posters

Extrapolation of polaron properties to low phonon frequencies by multi-fidelity

Bayesian machine learning

Vancouver, Canada

CONFERENCE TALK/JOURNAL ARTICLE

August 2023

- Presented a talk and poster at the RTGdyn Conference 2022.
- Kairon, P., Sous, J., Berciu, M., & Krems, R. V. (2024). Extrapolation of polaron properties to low phonon frequencies by Bayesian machine learning. *Physical Review B*, 109(14), 144523.

Equivalence between exponential concentration in quantum machine learning kernels and barren plateaus in variational algorithms

Vancouver, Canada

JOURNAL ARTICLE

August 2023

- <https://arxiv.org/abs/2501.07433v3> (to be published in PRA)
- Presented a talk and poster at the RTGdyn Conference 2024.

Coherence-based inequality for the discrimination of three-qubit GHZ and W class

Delhi, India

JOURNAL ARTICLE

January 2022

- Kairon, P., Singh, M., Adhikari, S. Coherence-based inequality for the discrimination of three-qubit GHZ and W class. *Quantum Inf Process* 21, 173 (2022)

Noisy three-player dilemma game: robustness of the quantum advantage

Delhi, India

JOURNAL ARTICLE

April 2020

- Kairon, P., Thapliyal, K., Srikanth, R. et al. Noisy three-player dilemma game: robustness of the quantum advantage. *Quantum Inf Process* 19, 327 (2020).

COVID-19 Outbreak Prediction Using Quantum Neural Networks

DoSIER 2020 (Virtual)

CONFERENCE PAPER

June 2021

- Kairon, P., Bhattacharyya, S. (2021). COVID-19 Outbreak Prediction Using Quantum Neural Networks. In: Bhattacharyya, S., Dutta, P., Datta, K. (eds) *Intelligence Enabled Research. Advances in Intelligent Systems and Computing*, vol 1279. Springer, Singapore.

- Presented an abstract on the simulation of envelope formation for SARS-CoV-2 to identify potential therapeutic targets. [Link](#)

Work Experience

Mitacs Accelerate

Canada

RESEARCH INTERN

October 2023 – August 2024

- Using quantum machine learning methods to perform feature selection for a metabolomics dataset with millions of dimensions.
- Collaborated on a grant proposal with a supervisor and a company, securing \$30k CAD in funding.

Superconducting Circuit Design and Fabrication by CMC Microsystems

Canada

WORKSHOP PARTICIPANT

June 2022 – July 2022

- Designed, simulated, and fabricated superconducting circuits, including resonators, SQUIDs, and qubits.
- Conducted advanced testing of superconducting chips using a 4K probe station and a dilution refrigerator.
- Gained hands-on experience with low-temperature experimental techniques for quantum information science.

HCL Technologies

Bengaluru, India

RESEARCH INTERN

August 2021 – September 2021

- Benchmarked methods for solving the electronic structure problem on DWave quantum annealer (AWS Amazon Braket).
- Explored error mitigation schemes for variational quantum eigensolvers using Hamiltonian symmetries.

Conduit Computing

Boston, USA (Remote)

RESEARCH INTERN

March 2020 – January 2021

- Simulated proteins for COVID-19 drug development using NAMD and VMD on the Frontera supercomputer.
- Proposed research selected for the White House HPC consortium and Nvidia Supercomputers.
- Presented a full research proposal at Quantum British Columbia in a virtual poster session.

Qnu Labs (India's First Quantum Company)

Bengaluru, India

RESEARCH INTERN

December 2019 – January 2020

- Developed MATLAB/Mathematica programs to calculate key rate deterioration with distance for DPS QKD protocols.
- Simulated high-throughput quantum random number generators based on LASER phase noise.

Solid State Physics Laboratory – DRDO

Delhi, India

RESEARCH INTERN

May 2019 – August 2019

- Characterized materials using Scanning Tunnelling Microscopy and fabricated MIS structures for AlN thin films.
- Repaired a table top STM, reconfiguring software and hardware setups independently.

Solid State Physics Laboratory – DRDO

Delhi, India

RESEARCH INTERN

December 2018 – February 2019

- Modeled thin film characterization using X-ray reflectivity with MATLAB.
- Determined composition, stress, and strain of AlGaIn/GaN multilayer structures for HEMT devices.

Laser Science and Technology Center – DRDO

Delhi, India

RESEARCH INTERN

May 2018 – August 2018

- Developed an optical resonator for high-power lasers using Nd-doped PoCl_3 as the gain medium.
- Surveyed various optical coatings for laser-grade mirrors.

Skills

Programming Languages	Python, C++, MATLAB, Mathematica, R
Packages and Frameworks	Qiskit, PennyLane, IBMQ Experience, JAX, TensorFlow, PyTorch, ITensor, dmrgpp
Utilities and Tools	Git/GitHub, Linux Shell, VSCode, Jupyter Notebook, SQL, MongoDB

Extracurricular Activities

NSERC Quantum Computing CREATE

Vancouver, Canada

MENTOR AND JUDGE

November 2022

- Mentored two groups of high school students on using quantum computing for interplanetary communication.
- Judged final presentations for the high school engineering and design competition.

- Organized seminars, debates, guest lectures, and paper presentations to enhance students' understanding of various scientific fields.
- Conducted lectures on advanced topics not covered in university courses and mentored beginners in systematic exploration of these topics.

Student Mentorship Programme

- Mentored five freshmen from the Engineering Physics Department.
- Provided guidance on both personal and academic challenges throughout the year.

Honors, Grants, & Scholarships

2023	Mitacs Accelerate Grant , Secured funding for a 10-month internship in collaboration with a biomedical startup to explore quantum machine learning applications.	Canada
2022	IBM Error Correction Summer School , Selected for in-person IBM summer school in New York from hundreds of applicants.	New York, USA
2022	Commendable Research Award at DTU , Received the award and a cash prize of Rs. 50k for publishing a journal article during my bachelor's thesis.	New Delhi, India
2022	Quantum BC Roadmapping Workshop Winner , Won first place in a workshop developing a quantum-based solution for an industrially relevant genomics problem.	Canada
2021	QuEST Fellowship , Awarded by the Stewart Blusson Quantum Matter Institute, University of British Columbia.	Vancouver, Canada
2021	NSERC Quantum Computing CREATE Scholarship , Received prestigious funding for research at UBC.	Vancouver, Canada
2021	MCQST Summer School Selection , Shortlisted among 20 students out of 500 applicants for the Munich Center of Quantum Science and Technology Summer School.	Munich, Germany (Virtual)
2020	Cornerstone Models of Quantum Computing Summer School , Selected for the summer school conducted by Quantum British Columbia, among 130 undergraduate and graduate candidates.	Canada (Virtual)
2017	All India Rank - 4922, JEE Main , Achieved this rank in the JEE Main examination, taken by approximately 1.5 million students.	India

References

1	Prof. Bryan K. Clark , University of Illinois Urbana Champaign, USA	Email
2	Prof. Roman Krems , University of British Columbia, Canada	Email
3	Prof. Anirban Pathak , Jaypee Institute of Information Technology, India	Email